```
!pip install transformers -q

import torch
from PIL import Image
from transformers import CLIPProcessor, CLIPModel
from google.colab import files
import io
import matplotlib.pyplot as plt

# Device setup
device = "cuda" if torch.cuda.is_available() else "cpu"

# Load CLIP model
model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32").to(device)
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")

# Baby body parts
body_parts = ["face", "hand", "arm", "leg", "foot", "torso"]
candidate_labels = [f"a photo of a baby {part}" for part in body_parts]

# Upload image
print("Upload a baby body-part image")
uploaded = files.upload()

if uploaded:
    filename = list(uploaded.keys())[0]
    image = Image.open(io.BytesIO(uploaded[filename])).convert("RGB")

    # CLIP prediction
    inputs = processor(
        text=candidate_labels,
        images=image,
        return_tensors="pt",
        padding=True
    ).to(device)

    with torch.no_grad():
        outputs = model(**inputs)

    probs = outputs.logits_per_image.softmax(dim=1)[0]

    # Top-2 predictions
    top_probs, top_idxs = torch.topk(probs, k=2)

    # Display image
    plt.imshow(image)
    plt.axis("off")
    plt.show()
```

```
    # Print predictions
    print("Predictions:")
    for i in range(2):
        label = body_parts[top_idxs[i]].upper()
        confidence = top_probs[i].item() * 100
        print(f"{i+1}. {label} — {confidence:.2f}%")

else:
    print("No image uploaded")
```

```
Loading weights: 100%                                                    398/398 [00:01<00:00, 319.32it/s, Materializing param=visual_projection.weight]
CLIPModel LOAD REPORT from: openai/clip-vit-base-patch32
Key                                 | Status     |  |
-----------------------------------+------------+--+-
vision_model.embeddings.position_ids | UNEXPECTED |  |
text_model.embeddings.position_ids   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Upload a baby body-part image
Choose Files foot.jpg
foot.jpg(image/jpeg) - 6140 bytes, last modified: 4/2/2026 - 100% done
Saving foot.jpg to foot.jpg
```

```
Predictions:
1. FOOT — 83.88%
2. HAND — 12.46%
```